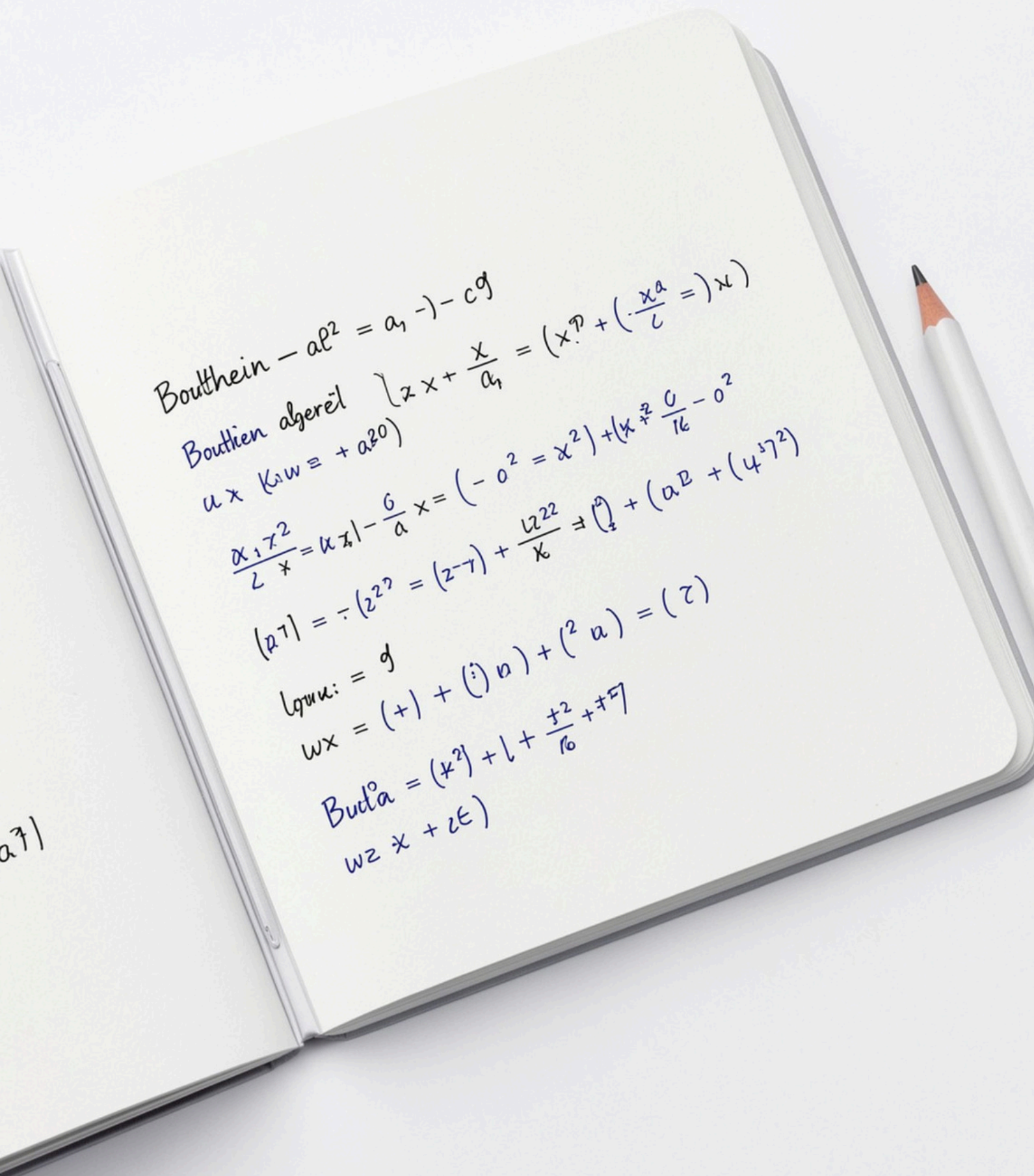




Sum of Products (SOP)

N.PRAVEEN KUMAR
26pca154

Sum of Products (SOP):

- Sum of Products (SOP) is a way of writing a Boolean expression using AND and OR operations.
- It is formed using Boolean variables and their complements.
- SOP can be obtained easily from a truth table by selecting the rows where the output is 1.
- It is commonly used to represent and design digital logic circuits.

Example:

For the values:

$$x_1 = 0, x_2 = 1, x_3 = 0,$$

The minterm is:

$$x_1' \ x_2 \ x_3'$$

If there are multiple rows with output 1, their minterms are added:

Minterm:

A minterm is a product (AND) of all Boolean variables.

- Each variable appears only once.
- If the variable value is 0 $\rightarrow$ take complement (').
- If the variable value is 1 $\rightarrow$ take the variable normally.

Variable	Value	Minterm
x_1	0	x_1'
x_2	1	x_2
x_3	0	x_3'
x_4	1	x_4
x_5	1	x_5

$$\text{Minterm} = x_1'x_2x_3'x_4x_5$$

Deriving SOP from Truth Table:

1

$$A = 0, B = 1$$

A is 0 $\rightarrow$ complement $\rightarrow A'$

B is 1 $\rightarrow$ keep $\rightarrow B$

Product term: $A'B$

2

$$A = 1, B = 0$$

A is 1 $\rightarrow$ keep $\rightarrow A$

B is 0 $\rightarrow$ complement $\rightarrow B'$

Product term: AB'

3

$$A = 1, B = 1$$

A is 1 $\rightarrow$ keep $\rightarrow A$

B is 1 $\rightarrow$ keep $\rightarrow B$

Product term: AB

Therefore: $F = A'B + AB' + AB$

SOP EXAMPLE:

$F(x,y,z)=(x+y)\bar{z}$

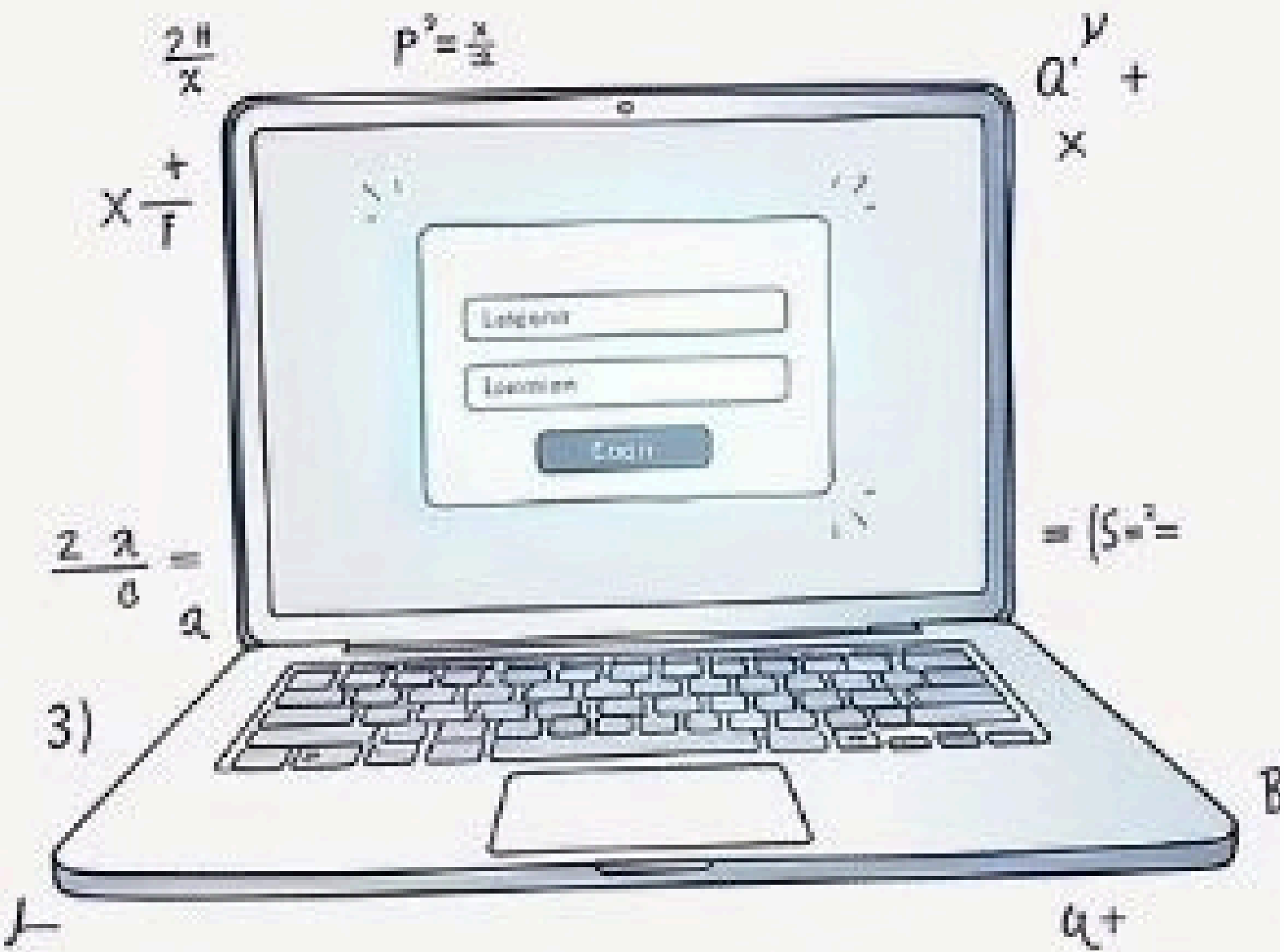
x	y	z	x + y	$\bar{z}$	F(x,y,z)
1	1	1	1	0	0
1	1	0	1	1	1
1	0	1	1	0	0
1	0	0	1	1	1
0	1	1	1	0	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	0	1	0

For F = 1: (x,y,z) = (1,0,0), (0,1,0)

$(SOP)F = xy'z' + x'yz'$

SOP in Computer Field:

- If $A = 1 \rightarrow$ Computer is ON
- If $B = 1 \rightarrow$ User is logged in
- Only when $A = 1$ and $B = 1$, then $F = 1$
- Therefore, Access is granted only when both conditions are true.



computer ON	Value	Result
0	0	NO ACCESS
0	1	NO ACCESS
1	0	NO ACCESS
1	1	Access Granted



Thank
you

